

Shicheng Fan

✉ sfan25@uic.edu 🌐 shichengf.github.io 📄 Google Scholar in shichengf0222

Research interests: **Causality** · **World Models** · **LLM Agents** · **LLM Post-training**

EDUCATION

University of Illinois Chicago Ph.D. in Computer Science *Aug 2025 – Present*
 Advised by [Prof. Philip S. Yu](#); collaborating with [Prof. Kun Zhang](#) (CMU).
Zhejiang University B.Eng. in Automation (Control & Robotics) *Sep 2021 – Jun 2025*
[Chu Kochen Honors College](#); GPA: **3.98**/4.0; Rank: **6/121 (Top 5%)**; Outstanding Graduate. Undergraduate thesis with [Prof. Qi Ye](#).

PUBLICATIONS & PREPRINTS

(**Bold** = Shicheng Fan; * = equal contribution.)

First-author

Shicheng Fan, Kun Zhang, Lu Cheng. “TRACE: Trajectory Recovery for Continuous Mechanism Evolution in Causal Representation Learning.” [Paper](#) / [Code](#) *ICML 2026*

Shicheng Fan*, Haochang Hao*, Dehai Min*, Weihao Liu, Philip S. Yu, Lu Cheng. “Verifiable Rewards Beyond Math and Code: Lightweight Corpus-Grounded Process Supervision for Factual Question Answering.” (**CorVer**) [Paper](#) / [Code](#) *Preprint*

Shicheng Fan, Nour Elhendawy, Jianle Sun, Ke Fang, Kun Zhang, Yihang Wang, Lu Cheng. “MOSAIC: Module Discovery via Sparse Additive Identifiable Causal Learning for Scientific Time Series.” [Paper](#) / [Code](#) *Preprint*

Collaborative publications

Hanrong Zhang*, **Shicheng Fan***, Henry Peng Zou, Yankai Chen, Zhenting Wang, Jiayu Zhou, Chengze Li, Wei-Chieh Huang, Yifei Yao, Kening Zheng, Xue Liu, Xiaoxiao Li, Philip S. Yu. “CoEvoSkills: Self-Evolving Agent Skills via Co-Evolutionary Verification.” [Paper](#) / [Code](#) *COLM 2026*

Hanrong Zhang, Yankai Chen, **Shicheng Fan**, Dehai Min, Shaowen Chen, ..., Dawn Song, Philip S. Yu, Xue Liu. “Scaling LLM Agent Learning with Data Synthesis: A Comprehensive Survey.” [Paper](#) *Preprint*

SELECTED RESEARCH PROJECTS

TRACE: Continuous Mechanism Evolution in Causal Representation Learning

First Author; ICML 2026

- Framed continuous mechanism evolution as a causal representation learning problem and proved identifiability through **nonlinear ICA** and **variational inference**.
- Built a **Mixture-of-Experts** model that recovers latent mechanism trajectories through time-varying mixture coefficients and interpolates to unseen states.

CoEvoSkills: Self-Evolving Agent Skills via Co-Evolutionary Verification

Co-first Author; COLM 2026

- Built a co-evolutionary framework for autonomous multi-file **agent skill** construction, pairing a Skill Generator with a Surrogate Verifier for actionable feedback.
- Evaluated self-evolving skills on **SkillsBench** across Claude Code, Codex, and additional LLM agents to analyze transfer and verifier reliability.

MOSAIC: Sparse Additive Identifiable Causal Learning for Scientific Time Series

First Author; Preprint

- Designed a **sparse temporal VAE** that identifies latent mechanisms from regime-conditioned temporal variation and maps each latent to sparse observed-variable supports.
- Established support identifiability for additive observation models and validated on RNA dynamics, solar wind, ENSO climate, Tennessee Eastman, and tokamak data.

CorVer: Corpus-Grounded Process Supervision for Factual QA

Co-first Author; Preprint

- Developed lightweight **corpus-grounded process supervision** that provides verifiable rewards for factual question answering beyond math and code tasks.
- Integrated retrieval-grounded reward signals into an **LLM post-training** pipeline combining SFT and RL for factuality alignment.

Semantic and Goal-Driven Dexterous Hand Manipulation

B.Eng. Thesis; 2024 – 2025

- Reproduced the **Text2HOI** three-stage framework, matching reported performance on the **H2O**, **GRAB**, and **ARCTIC** datasets.
- Proposed a **Shadow Hand** → **MANO** cross-embodiment retargeting algorithm; joint training with **GraspM3** improved Physical Realism to **0.897** and reduced penetration volume by **~50%**.

AWARDS & HONORS

Outstanding Graduate, Zhejiang University	<i>2025</i>
Academic Excellence Model, Zhejiang University	<i>2022, 2023, 2024</i>
Outstanding Student in AI&ML Summer Project, University of Cambridge	<i>2023</i>
National Second Prize, Advanced Mathematics Competition of China	<i>2022</i>

COMPETITIONS

ASC Student Supercomputer Challenge	<i>National Second Prize; 2024</i>
Built an end-to-end HPC system and optimized LLM inference with KV-cache and batching strategies.	
ACM-IPCC International Parallel Computing Challenge	<i>National Second Prize; 2022</i>
Reduced program runtime from 13 minutes to 200 ms using MPI, CUDA, and SIMD.	